

Bank Loan Report Query Document

A. Bank Loan Report | Summary

KPI's:

Total Loan Applications

Select Count (id) AS Total_Loan_Applications FROM bank_loan_data

	Total_Loan_Applications
1	38576

MTD Loan Application

Select Count (id) AS MTD_Total_Loan_Applications FROM bank_loan_data
WHERE MONTH (issue_date) = 12 AND YEAR(issue_date) = 2021

	MTD_Total_Loan_Applications
1	4314

PMTD Loan Application

SELECT Count (id) AS PMTD_Total_Loan_Applications FROM bank_loan_data
WHERE MONTH (issue_date) = 11 AND YEAR(issue_date) = 2021

	PMTD_Total_Loan_Applications
1	4035

Total Funded Amount

SELECT SUM (loan_amount) AS Total_Funded_Amount FROM bank_loan_data

	Total_Funded_Amount
1	435757075

MTD Total Funded Amount

```
SELECT SUM(loan_amount) AS MTD_Total_Funded_Amount FROM bank_loan_data  
WHERE MONTH(issue_date) = 12 AND YEAR(issue_date) = 2021
```

	MTD_Total_Funded_Amount
1	53981425

PMTD Total Funded Amount

```
SELECT SUM(loan_amount) AS PMTD_Total_Funded_Amount FROM bank_loan_data  
WHERE MONTH(issue_date) = 11 AND YEAR(issue_date) = 2021
```

	PMTD_Total_Funded_Amount
1	47754825

Total Amount Received

```
SELECT SUM(total_payment) AS Total_Amount_Received FROM bank_loan_data
```

	Total_Amount_Received
1	473070933

MTD Total Amount Received

```
SELECT SUM(total_payment) AS MTD_Total_Amount_Received FROM bank_loan_data  
WHERE MONTH(issue_date) = 12 AND YEAR(issue_date) = 2021
```

	MTD_Total_Amount_Received
1	58074380

PMTD Total Amount Received

```
SELECT SUM(total_payment) AS PMTD_Total_Amount_Received FROM bank_loan_data  
WHERE MONTH(issue_date) = 11 AND YEAR(issue_date) = 2021
```

	PMTD_Total_Amount_Received
1	50132030

Average Interest Rate

```
SELECT ROUND(AVG(int_rate), 4) * 100 AS Avg_Interest_Rate FROM bank_loan_data
```

	Avg_Interest_Rate
1	12.05

MTD Average Interest Rate

```
SELECT ROUND(AVG(int_rate), 4) * 100 AS MTD_Avg_Interest_Rate FROM  
bank_loan_data  
WHERE MONTH(issue_date) = 12 AND YEAR(issue_date) = 2021
```

	MTD_Avg_Interest_Rate
1	12.36

PMTD Average Interest Rate

```
SELECT ROUND(AVG(int_rate), 4) * 100 AS PMTD_Avg_Interest_Rate FROM  
bank_loan_data  
WHERE MONTH(issue_date) = 11 AND YEAR(issue_date) = 2021
```

	PMTD_Avg_Interest_Rate
1	11.94

Average Debt to Income Ratio (DTI)

```
SELECT ROUND(AVG(dti),4) *100 AS Avg_DTI FROM bank_loan_data
```

	Avg_DTI
1	13.33

MTD Average Debt to Income Ratio (DTI)

```
SELECT ROUND(AVG(dti),4) *100 AS MTD_Avg_DTI FROM bank_loan_data  
WHERE MONTH(issue_date) = 12 AND YEAR(issue_date) = 2021
```

	MTD_Avg_DTI
1	13.67

PMTD Average Debt to Income Ratio (DTI)

```
SELECT ROUND(AVG(dti),4) *100 AS PMTD_Avg_DTI FROM bank_loan_data  
WHERE MONTH(issue_date) = 11 AND YEAR(issue_date) = 2021
```

	PMTD_Avg_DTI
1	13.3

GOOD LOAN ISSUED:

Good Loan Percentage

```
SELECT  
(COUNT (CASE WHEN loan_status = 'Fully Paid' OR loan_status = 'Current'  
THEN id END))  
/  
COUNT (id) AS Good_loan_percentage  
FROM bank_loan_data
```

	Good_loan_percentage
1	86

Good Loan Applications

```
SELECT COUNT (id) AS Good_Loan_Applications FROM bank_loan_data
WHERE loan_status = 'Fully Paid' OR loan_status = 'Current'
```

	Good_Loan_Applications
1	33243

Good Loan Funded Amount

```
SELECT SUM (loan_amount) AS Good_Loan_Funded_Amount FROM bank_loan_data
WHERE loan_status = 'Fully Paid' OR loan_status = 'Current'
```

	Good_Loan_Funded_Amount
1	370224850

Good Loan Total Received Amount

```
SELECT SUM (total_payment) AS Good_Loan_Received_Amount FROM
bank_loan_data
WHERE loan_status = 'Fully Paid' OR loan_status = 'Current'
```

	Good_Loan_Received_Amount
1	435786170

Bad Loan Issued:

Bad Loan Percentage

```
SELECT
(COUNT (CASE WHEN loan_status = 'Charged off' THEN id END) *
100) /
COUNT (id) AS Bad_Loan_Percentage
FROM bank_loan_data
```

	Bad_Loan_Percentage
1	13

Bad Loan Applications

```
SELECT COUNT (id) AS Bad_Loan_Applications FROM bank_loan_data
WHERE loan_status = 'Charged off'
```

	Bad_Loan_Applications
1	5333

Bad Loan Funded Amount

```
SELECT SUM (loan_amount) AS Bad_Loan_Funded_Amount FROM bank_loan_data
WHERE loan_status = 'Charged off'
```

	Bad_Loan_Funded_Amount
1	65532225

Bad Loan Total Amount Received

```
SELECT SUM (total_payment) AS Bad_Loan_Received_Amount FROM
bank_loan_data
WHERE loan_status = 'Charged off'
```

	Bad_Loan_Received_Amount
1	37284763

Loan Status:

```
SELECT
    loan_status,
    Count(id) AS Total_Loan_applications,
    SUM(total_payment) AS Total_Amount_Received,
    SUM(loan_amount) AS Total_Funded_Amount,
    AVG(int_rate * 100) AS Interest_Rate,
    AVG(dti * 100) AS DTI
FROM bank_loan_data
GROUP BY loan_status
```

	loan_status	Total_Loan_applications	Total_Amount_Received	Total_Funded_Amount	Interest_Rate	DTI
1	Fully Paid	32145	411586256	351358350	11.6410707918092	13.1673507557434
2	Charged Off	5333	37284763	65532225	13.8785749318289	14.0047328005517
3	Current	1098	24199914	18866500	15.0993260800947	14.7243442736843

SELECT

loan_status,

SUM(total_payment) AS MTD_Total_Amount_Received,

SUM(loan_amount) AS MTD_Total_Funded_Amount

FROM bank_loan_data

WHERE MONTH(issue_date) = 12

GROUP BY loan_status

	loan_status	MTD_Total_Amount_Received	MTD_Total_Funded_Amount
1	Fully Paid	47815851	41302025
2	Charged Off	5324211	8732775
3	Current	4934318	3946625

B. BANK LOAN REPORT | OVERVIEW/CHARTS

MONTH:

SELECT

MONTH(issue_date) AS Month_Number,

DATENAME(MONTH, issue_date) AS

Month_Name, COUNT(id) AS

Total_Loan_Applications,

SUM(loan_amount) AS Total_Funded_Amount,

SUM(total_payment) AS Total_Amount_Received

FROM bank_loan_data

GROUP BY MONTH(issue_date), DATENAME(MONTH, issue_date)

ORDER BY MONTH(issue_date)

	Month_Number	Month_Name	Total_Loan_Applications	Total_Funded_Amount	Total_Amount_Received
1	1	January	2332	25031650	27578836
2	2	February	2279	24647825	27717745
3	3	March	2627	28875700	32264400
4	4	April	2755	29800800	32495533
5	5	May	2911	31738350	33750523
6	6	June	3184	34161475	36164533
7	7	July	3366	35813900	38827220
8	8	August	3441	38149600	42682218
9	9	September	3536	40907725	43983948
10	10	October	3796	44893800	49399567
11	11	November	4035	47754825	50132030
12	12	December	4314	53981425	58074380

STATE:

SELECT

Address_state AS State,

COUNT(id) AS Total_Loan_Applications,

SUM(loan_amount) AS Total_Funded_Amount,

SUM(total_payment) AS Total_Amount_Received

FROM bank_loan_data

GROUP BY address_state

ORDER BY address_state

	State	Total_Loan_Applications	Total_Funded_Amount	Total_Amount_Received
1	AK	78	1031800	1108570
2	AL	432	4949225	5492272
3	AR	236	2529700	2777875
4	AZ	833	9206000	10041986
5	CA	6894	78484125	83901234
6	CO	770	8976000	9845810
7	CT	730	8435575	9357612
8	DC	214	2652350	2921854
9	DE	110	1138100	1269136
10	FL	2773	30046125	31601905
11	GA	1355	15480325	16728040
12	HI	170	1850525	2080184
13	IA	5	56450	64482
14	ID	6	59750	65329
15	IL	1486	17124225	18875941
16	IN	9	86225	85521
17	KS	260	2872325	3247394
18	KY	320	3504100	3792530
19	LA	426	4498900	5001160
20	MA	1310	15051000	16676279
21	MD	1027	11911400	12985170
22	ME	3	9200	10808
23	MI	685	7829900	8543660
24	MN	592	6302600	6750746
25	MO	660	7151175	7692732

TERM:

SELECT

term AS Term,
COUNT(id) AS Total_Loan_Applications,
SUM(loan_amount) AS Total_Funded_Amount,
SUM(total_payment) AS Total_Amount_Received

FROM bank_loan_data

GROUP BY term

ORDER BY term

	Term	Total_Loan_Applications	Total_Funded_Amount	Total_Amount_Received
1	36 months	28237	273041225	294709458
2	60 months	10339	162715850	178361475

EMPLOYEE LENGTH:

SELECT

emp_length AS Employee_Length,
COUNT(id) AS Total_Loan_Applications,
SUM(loan_amount) AS Total_Funded_Amount,
SUM(total_payment) AS Total_Amount_Received
FROM bank_loan_data
GROUP BY emp_length
ORDER BY emp_length

	Employee_Length	Total_Loan_Applications	Total_Funded_Amount	Total_Amount_Received
1	< 1 year	4575	44210625	47545011
2	1 year	3229	32883125	35498348
3	10+ years	8870	116115950	125871616
4	2 years	4382	44967975	49206961
5	3 years	4088	43937850	47551832
6	4 years	3428	37600375	40964850
7	5 years	3273	36973625	40397571
8	6 years	2228	25612650	27908658
9	7 years	1772	20811725	22584136
10	8 years	1476	17558950	19025777
11	9 years	1255	15084225	16516173

Purpose:

SELECT

```
    purpose AS PURPOSE,  
    COUNT(id) AS Total_Loan_Applications,  
    SUM(loan_amount) AS Total_Funded_Amount,  
    SUM(total_payment) AS Total_Amount_Received  
FROM bank_loan_data  
GROUP BY purpose  
ORDER BY purpose
```

	PURPOSE	Total_Loan_Applications	Total_Funded_Amount	Total_Amount_Received
1	car	1497	10223575	11324914
2	credit card	4998	58885175	65214084
3	Debt consolidation	18214	232459675	253801871
4	educational	315	2161650	2248380
5	home improvement	2876	33350775	36380930
6	house	366	4824925	5185538
7	major purchase	2110	17251600	18676927
8	medical	667	5533225	5851372
9	moving	559	3748125	3999899
10	other	3824	31155750	33289676
11	renewable_energy	94	845750	898931
12	small business	1776	24123100	23814817
13	vacation	352	1967950	2116738
14	wedding	928	9225800	10266856

HOME OWNERSHIP:

SELECT

```
    home_ownership AS Home_Ownership,  
    COUNT(id) AS Total_Loan_Applications,  
    SUM(loan_amount) AS Total_Funded_Amount,  
    SUM(total_payment) AS Total_Amount_Received  
FROM bank_loan_data  
GROUP BY home_ownership  
ORDER BY home_ownership
```

	Home_Ownership	Total_Loan_Applications	Total_Funded_Amount	Total_Amount_Received
1	MORTGAGE	17198	219329150	238474438
2	NONE	3	16800	19053
3	OTHER	98	1044975	1025257
4	OWN	2838	29597675	31729129
5	RENT	18439	185768475	201823056

Note: I have applied multiple filters on all the dashboards. You can check the results for the filters as well by modifying the query and comparing the results.

For e.g.

See the results when we hit the grade A in the filters for dashboards

```
SELECT
    purpose AS PURPOSE,
    COUNT(id) AS Total_Loan_Applications,
    SUM(loan_amount) AS Total_Funded_Amount,
    SUM(total_payment) AS Total_Amount_Received
FROM bank_loan_data
WHERE grade = 'A'
GROUP BY purpose
ORDER BY purpose
```